

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q22-2020

Oxford Dam 138 kV
January 30, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2020. The Interconnection Customer proposes a 41.0 MW natural gas combined cycle facility interconnecting at the Oxford Dam 138 kV substation in the Virginia area. The request was received on February 28, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 495 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,732,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Dps Ocean Spray / Windsor Hills 138 kV	Short Pump / S Meadow 2 138 kV	133.2	230	6.4
Southwick / Windsor Hills 138 kV	Short Pump / S Meadow 2 138 kV	131.2	260	6.4
Dps Ocean Spray / Pipers Gap 138 kV	Birchwood / Leavitt 345 kV	116.8	229	6.0
Pipers Gap / Mesa 138 kV	Leavitt Substation / Birchwood 345 kV L	116.7	257	6.0
Short Pump / Mesa 138 kV L	Leavitt / Birchwood 345 kV L	116.6	288	6.0
Short Pump / S Meadow 2 138 kV line	Birchwood / Leavitt 345 kV Line	116.4	359	6.0
Dps Ocean Spray / Four Rivers 138 kV	Leavitt / Birchwood 345 kV Line	103.4	466	7.7
Four Rivers / Trenton Diesel 138 kV	Leavitt / Birchwood 345 kV line	103.4	466	7.7
Nestuca / Polk City 345 kV	Birchwood / Nauvoo 345 kV	102.7	1,458	5.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Dps Ocean Spray / Windsor Hills 138 kV	Rebuild line with higher-capacity conductor	\$5,685,000	10% (\$573,000)
Southwick / Windsor Hills 138 kV	Rebuild line with higher-capacity conductor	\$6,757,000	10% (\$681,000)
Dps Ocean Spray / Pipers Gap 138 kV	Rebuild line with higher-capacity conductor	\$3,251,000	10% (\$324,000)
Pipers Gap / Mesa 138 kV	Rebuild line with higher-capacity conductor	\$2,701,000	10% (\$269,000)
Short Pump / Mesa 138 kV L	Rebuild line with higher-capacity conductor	\$3,005,000	10% (\$300,000)
S Meadow 2 / Short Pump 138 kV L	Rebuild line with higher-capacity conductor	\$3,923,000	10% (\$391,000)
Dps Ocean Spray / Four Rivers 138 kV	Reconductor line	\$2,314,000	2% (\$55,000)
Trenton Diesel / Four Rivers 138 kV	Reconductor line	\$4,897,000	2% (\$117,000)
Nestuca / Polk City 345 kV	Reconductor line	\$5,710,000	0% (\$22,000)
	Total allocated to project		\$2,732,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.